A PROJECT REPORT

FAKE JOB POSTING DETECTION

Submitted by

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to

the O.P. Jindal University

in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Science (Hons.)

in

Data Science and Analytics

Department of Mathematics

O.P. Jindal University, Raigarh, 496001

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CERTIFICATE

This is to certify that the project work entitled "**Fake Job Posting Detection System Using Machine Learning**" has been successfully carried out by **Mr. Digreelal Baiga, Mr. Parth Ghosh, Mr. Bukesh Kumar Banpela, Mr. Harshdeep Toppo, Mr. Minesh Darshan** under my supervision.

This work is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Science (Honours) in Data Science & Analytics** in the School of Science, **O P Jindal University, Raigarh (C.G.)**.

To the best of my knowledge, the work presented by the student is genuine and has been completed satisfactorily. The project is found to be complete in all respects and is ready for submission to the University.

Place : OPJU, Raigarh, (C.G.)

Mr. Jitendra Kumar Gardia
(Project Guide)
Department of CSE, SOE
O.P. Jindal University Raigarh

DECLARATION

We hereby declare that the project report entitled "**Fake Job Posting Detection System Using Machine Learning**" is a genuine and original work carried out by us as part of the requirements for the award of the **Degree of Bachelor of Science (Honours) in Data Science & Analytics** at **O.P. Jindal University, Raigarh (Chhattisgarh)**, under the guidence of **Asst. Prof. Mr. Jitendra Kumar Gardia.**

This project has been completed under the valuable guidance and supervision of our mentor. We confirm that the work presented in this report is entirely our own effort and has not been copied or reproduced from any other source, except where proper acknowledgment and references have been provided.

We further declare that this project report has not been submitted, either fully or partially, for the award of any other degree, diploma, or certification in any institution. All the data, analysis, results, and conclusions presented in this report are based on our independent work and learning during the project period.

We take full responsibility for the authenticity, originality, and accuracy of the content presented in this report. Any errors or omissions, if found, are solely our responsibility.

Certified that the above statement made by the student is correct to the best of our knowledge and belief.

Examined By:
Prof. Manas Ranjan Mishra **Head of Department**

ACKNOWLEDGEMENT

We would like to express our sincere gratitude and heartfelt thanks to all those who have supported and guided us throughout the completion of our project titled "**Fake Job Posting Detection System Using Machine Learning.**"

First and foremost, We would like to thank our respected project guide, **Mr. Jitendra Kumar Gardia**, for his continuous guidance, valuable suggestions, and constant encouragement during the entire course of this project. His expertise, patience, and insightful feedback have played a crucial role in shaping this project successfully.

We extend our sincere gratitude to the **Department of Data Science & Analytics, O.P. Jindal University, Raigarh**, for providing the necessary academic environment, resources, and infrastructure required to complete this work efficiently.

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Last but not least, We would like to thank our family for their constant support, understanding, and encouragement, which gave me the strength and confidence to complete this project successfully.

This project has been a great learning experience for us, enhancing both our technical knowledge and problem-solving skills.

Place : OPJU, Raigarh, (C.G.)

Digreelal Baiga
Parth Ghosh
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TABLE OF CONTENTS

CHAPTER 1
BASIC THOUGHTS

1.1 Project Overview

Fake Job Posting Detection System Using Machine Learning

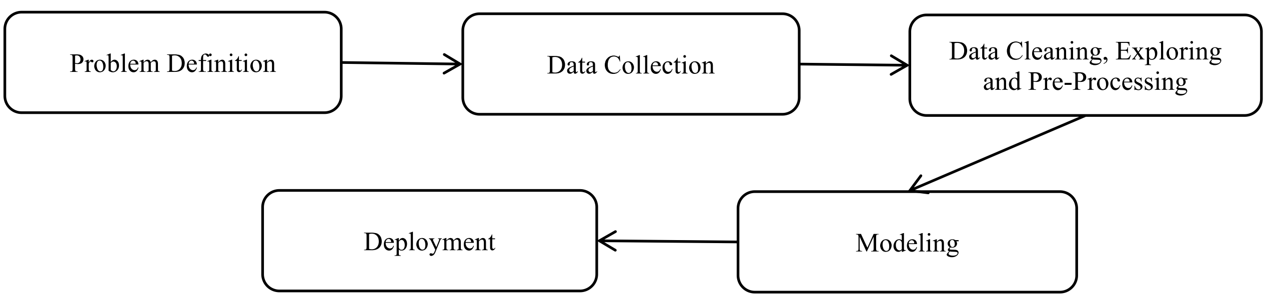
The Fake Job Posting Detection System is a machine learning–based project developed to identify fraudulent job advertisements posted on online job platforms. With the rapid growth of online recruitment websites, many scammers create fake job postings to deceive job seekers by offering unrealistic salaries, work-from-home opportunities, or asking for registration fees and personal information.

This project aims to build an automated system that can analyze job descriptions and detect whether a job posting is real or fake using Natural Language Processing (NLP) and Machine Learning techniques.

Employment scams are on the rise. According to CNBC, the number of employment scams doubled in 2018 as compared to 2017. The current market situation has led to high unemployment. Economic stress and the impact of the coronavirus have significantly reduced job availability and the loss of jobs for many individuals. A case like this presents an appropriate opportunity for scammers. Many people are falling prey to these scammers using the desperation that is caused by an unprecedented incident. Most scammer do this to get personal information from the person they are scamming. Personal information can contain address, bank account details, social security number etc. The scammers provide users with a very lucrative job opportunity and later ask for money in return. Or they require investment from the job seeker with the promise of a job. This is a dangerous problem that can be addressed through Machine Learning techniques and Natural Language Processing (NLP).

This project uses data provided from Kaggle. This data contains features that define a job posting. These job postings are categorized as either real or fake. Fake job postings are a very small fraction of this dataset. That is as excepted. We do not expect a lot of fake jobs postings. This project follows five stages.

The five stages adopted for this project are –
1. Problem Definition (Project Overview, Project statement and Metrics)
2. Data Collection
3. Data cleaning, Exploring and Pre-Processing
4. Modeling
5. Deployment



1.2 Abstract

Fake job postings are increasing rapidly on online job portals and social media platforms. Many fraudulent companies post fake job advertisements to collect personal information, charge fees, or scam job seekers. Detecting such fraudulent job postings manually is difficult due to the large number of postings available online.

This project aims to develop a machine learning model that can automatically detect fake job postings. The dataset used in this project contains job descriptions, company profiles, requirements, benefits, and other job-related information. Text preprocessing techniques such as cleaning, tokenization, and TF-IDF vectorization are applied to convert textual data into numerical form.

Machine learning algorithms such as Logistic Regression are used to train the model to classify job postings as real or fake. The trained model is evaluated using metrics such as accuracy, confusion matrix, and classification report.

Finally, a web application is developed using Streamlit where users can enter a job description and the system predicts whether the job posting is real or fraudulent.

1.3 Introduction

Online job portals such as LinkedIn, Indeed, and Glassdoor are widely used for job searching. However, these platforms are also exploited by scammers who post fake job advertisements to deceive job seekers.

Fake job postings may request personal information, charge application fees, or promise unrealistic salaries. These scams can result in financial loss and identity theft.

Therefore, there is a need for an automated system that can identify fraudulent job postings. Machine learning techniques can analyze job descriptions and identify patterns that distinguish real jobs from fake ones.

The objective of this project is to build a machine learning model that can detect fake job postings using natural language processing techniques.

1.4 Problem Statement

Many online job postings are fraudulent and difficult to identify manually. Job seekers may fall victim to scams due to misleading job descriptions.

The goal of this project is to develop a machine learning system that automatically detects whether a job posting is real or fake based on its textual content.

Metrics

The models will be evaluated based on four metrics:

(1). Precision

$$\text{Precision} = \frac{\text{True positive}}{\text{True Positive + False Positive}}$$

(2). Recall

$$\text{Recall}=\frac{\text{True Positive}}{\text{True positive + False Negative}}$$

(3). Accuracy

$$\text{Accuracy}=\frac{\text{True Positive + True Negativee}}{\text{True Positve + False Negative + True Negative + False Positive}}$$

(4). F1-Score

$$\text{F1}-\text{Score}=\frac{\text{True Positive}}{\text{True Positive} + \frac{1}{2}\ (\text{False Positive + False negative})}$$

1.5 Objectives

- To collect and analyze a dataset of job postings.
- To preprocess and clean textual data.
- To convert text data into numerical features using TF-IDF.
- To train machine learning models for fake job detection.
- To evaluate model performance using accuracy and other metrics.
- To build a user-friendly web application for prediction.

CHAPTER 2
DATASET DESCRIPTION AND DATA UNDERSTANDING

2.1 Dataset Description

The dataset used in this project is Fake Job Postings Dataset.

#	Column Name	Data Type	Description
1	job_id	Integer	Unique identification number assigned to each job posting.
2	title	Text	The title of the job position (e.g., Data Analyst, Software Engineer).
3	location	Text	The geographical location where the job is offered.
4	department	Text	Department within the organization where the job belongs.
5	salary_range	Text	The salary range offered for the job position.
6	company_profile	Text	Description of the company posting the job advertisement.
7	description	Text	Detailed information about job responsibilities and tasks.
8	requirements	Text	Skills, qualifications, and experience required for the job.
9	benefits	Text	Benefits provided by the company such as insurance, bonuses, etc.
10	telecommuting	Integer (0/1)	Indicates whether the job allows remote work (1 = Yes, 0 = No).
11	has_company_logo	Integer (0/1)	Indicates whether the company logo is included in the job posting.
12	has_questions	Integer (0/1)	Indicates whether screening questions are included in the job posting.
13	employment_type	Text	Type of employment such as Full-time, Part-time, Contract.
14	required_experience	Text	Experience level required for the job (Entry level, Mid level, etc.).
15	required_education	Text	Educational qualification required for the job.
16	industry	Text	Industry category of the job (IT, Marketing, Finance, etc.).
17	function	Text	Job function or role category.
18	fraudulent	Integer (0/1)	Target variable indicating whether the job posting is fake (1) or real (0).

The dataset used in this project contains multiple features related to job advertisements such as job title, company profile, job description, requirements, and benefits. These features provide valuable textual information that helps identify patterns commonly found in fake job postings. The target variable fraudulent indicates whether the job posting is real or fake. The dataset contains both textual and numerical features, which are processed using Natural Language Processing techniques and machine learning algorithms to detect fraudulent job advertisements.

Since most of the datatypes are either booleans or text a summary statistic is not needed here. The only integer is job_id which is not relevant for this analysis. The dataset is further explored to identify null values.

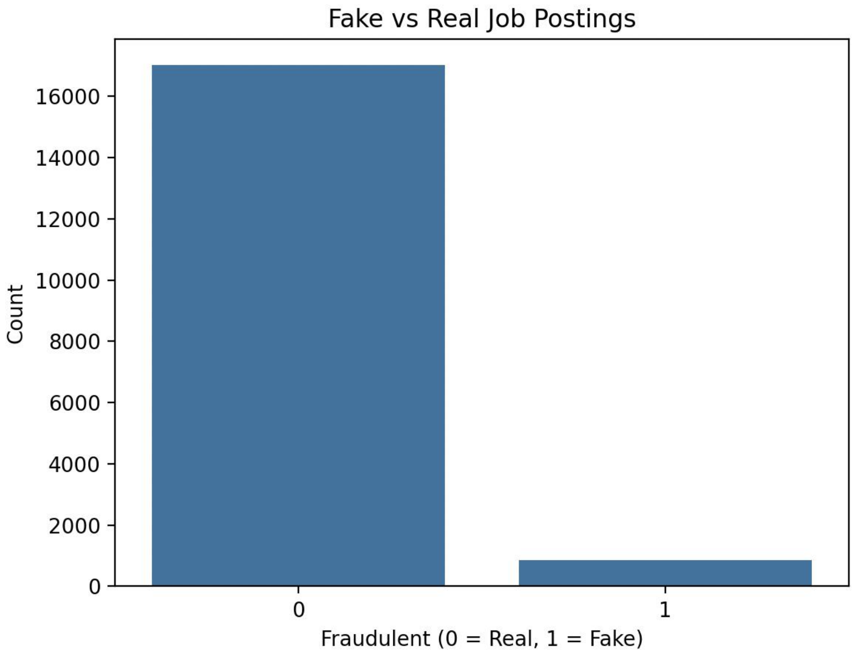
```
job_id                     0
title                      0
location                 346
department             11547
salary_range           15012
company_profile         3308
description                1
requirements            2696
benefits                7212
telecommuting              0
has_company_logo           0
has_questions              0
employment_type         3471
required_experience     7050
required_education      8105
industry                4903
function                6455
fraudulent                 0
dtype: int64
```

Variables such as department and salary_range have a lot of missing values. These columns are dropped from further analysis.

After initial assessment of the dataset, it could be seen that since these job postings have been extracted from several countries the postings were in different languages. To simplify the process this project uses data from US based locations that account for nearly 60% of the dataset. This was done to ensure all the data is in English for easy interpretability

Also, the location is split into state and city for further analysis. The final dataset has 17881 observations and 18 features.

The dataset is highly unbalanced with 16000 (93% of the jobs) being real and only 725 or 7% of the jobs being fraudulent. A countplot of the same can show the disparity very clearly.



2.2 Data Understanding

- Head of Dataset

```
   job_id                                      title  ...              function fraudulent
0       1                           Marketing Intern  ...             Marketing          0
1       2  Customer Service - Cloud Video Production  ...      Customer Service          0
2       3    Commissioning Machinery Assistant (CMA)  ...                   NaN          0
3       4          Account Executive - Washington DC  ...                 Sales          0
4       5                        Bill Review Manager  ...  Health Care Provider          0
```

- Shape of Dataset

(17880, 18)

- Column of Dataset

```
Index(['job_id', 'title', 'location', 'department', 'salary_range',
       'company_profile', 'description', 'requirements', 'benefits',
       'telecommuting', 'has_company_logo', 'has_questions', 'employment_type',
       'required_experience', 'required_education', 'industry', 'function',
       'fraudulent'],
      dtype='object')
```

➢ Information of Dataset

#	Column	Non-Null Count	Dtype
0	job_id	17880 non-null	int64
1	title	17880 non-null	object
2	location	17534 non-null	object
3	department	6333 non-null	object
4	salary_range	2868 non-null	object
5	company_profile	14572 non-null	object
6	description	17879 non-null	object
7	requirements	15184 non-null	object
8	benefits	10668 non-null	object
9	telecommuting	17880 non-null	int64
10	has_company_logo	17880 non-null	int64
11	has_questions	17880 non-null	int64
12	employment_type	14409 non-null	object
13	required_experience	10830 non-null	object
14	required_education	9775 non-null	object
15	industry	12977 non-null	object
16	function	11425 non-null	object
17	fraudulent	17880 non-null	int64

➢ Describe of Dataset

	job_id	telecommuting	has_company_logo	has_questions	fraudulent
count	17880.000000	17880.000000	17880.000000	17880.000000	17880.000000
mean	8940.500000	0.042897	0.795302	0.491723	0.048434
std	5161.655742	0.202631	0.403492	0.499945	0.214688
min	1.000000	0.000000	0.000000	0.000000	0.000000
25%	4470.750000	0.000000	1.000000	0.000000	0.000000
50%	8940.500000	0.000000	1.000000	0.000000	0.000000
75%	13410.250000	0.000000	1.000000	1.000000	0.000000
max	17880.000000	1.000000	1.000000	1.000000	1.000000

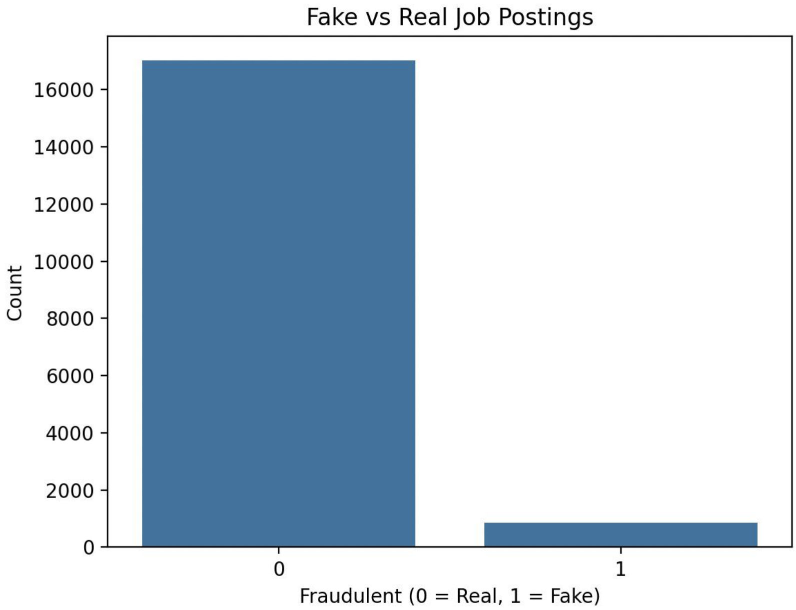
➢ Missing Values

job_id	0
title	0
location	346
department	11547
salary_range	15012
company_profile	3308
description	1
requirements	2696
benefits	7212
telecommuting	0
has_company_logo	0
has_questions	0
employment_type	3471
required_experience	7050
required_education	8105
industry	4903
function	6455
fraudulent	0

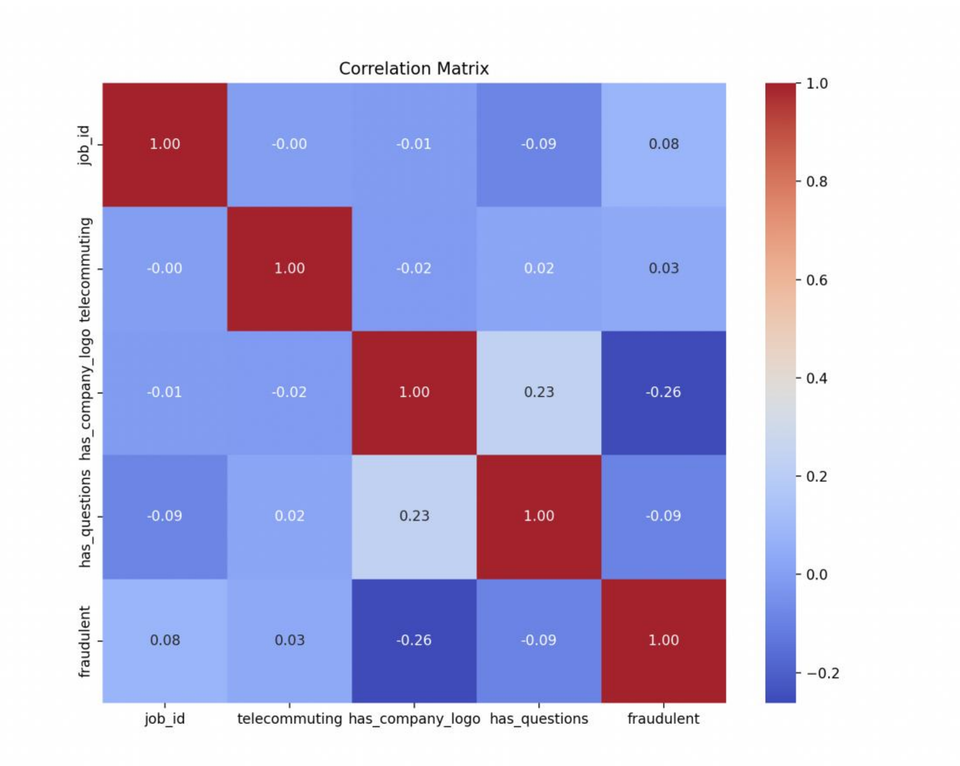
dtype: int64

CHAPTER 3
EXPLORATORY VISUALIZATION

The first step to visualize the dataset in this project is to create a count plot can show the disparity very clearly.



Visualize the dataset in this project is to create a correlation matrix to study the relationship between the numeric data.

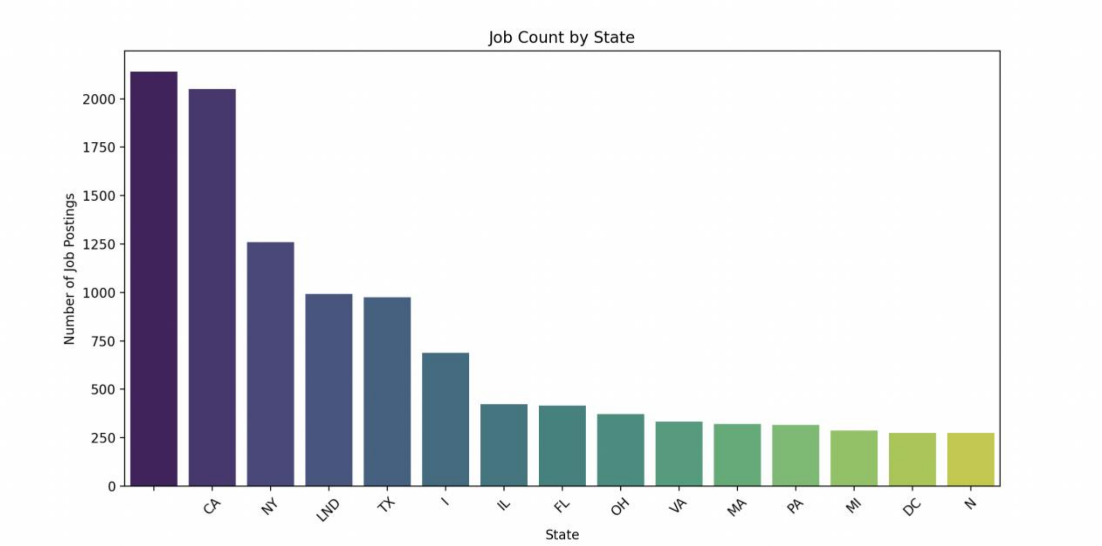


The

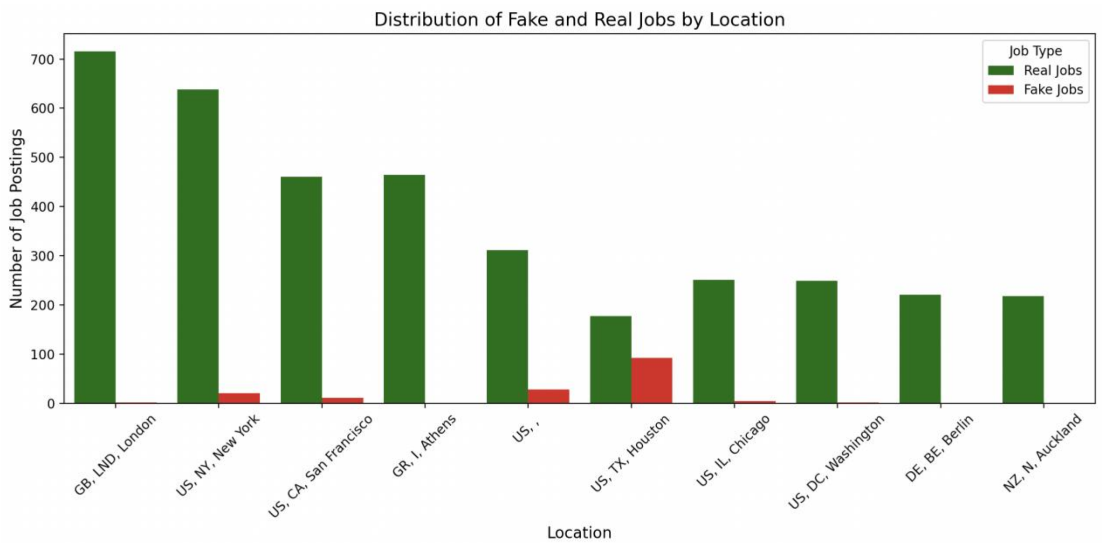
correlation matrix does not exhibit any strong positive or negative correlations between the numeric data.

However, an interesting trend was noted with respect to the Boolean variable *telecommuting*. In cases when both this variable had value equal to zero there is a 92% chance that the job will be fraudulent.

After the numeric features the textual features of this dataset is explored. We start this exploration from location.



The graph above shows which states produces the greatest number of jobs. California, New York and Texas have the highest number of job postings. To explore this further another bar plot is created. This barplot shows the distribution of fake and real jobs in the top 10 states.

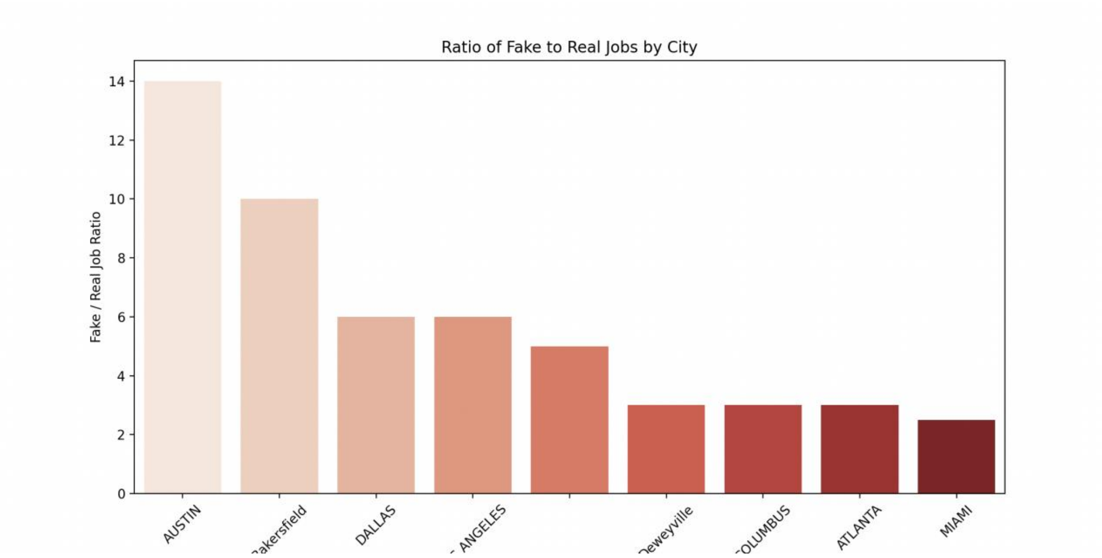


The graph above shows that Texas and California have a higher possibility of fake jobs as compared to other states.

To dig one level deeper into and include states as well a ratio is created. This is a fake to real job ratio based on states and citifies. The following formula is used to compute how many fake jobs are available for every real job.

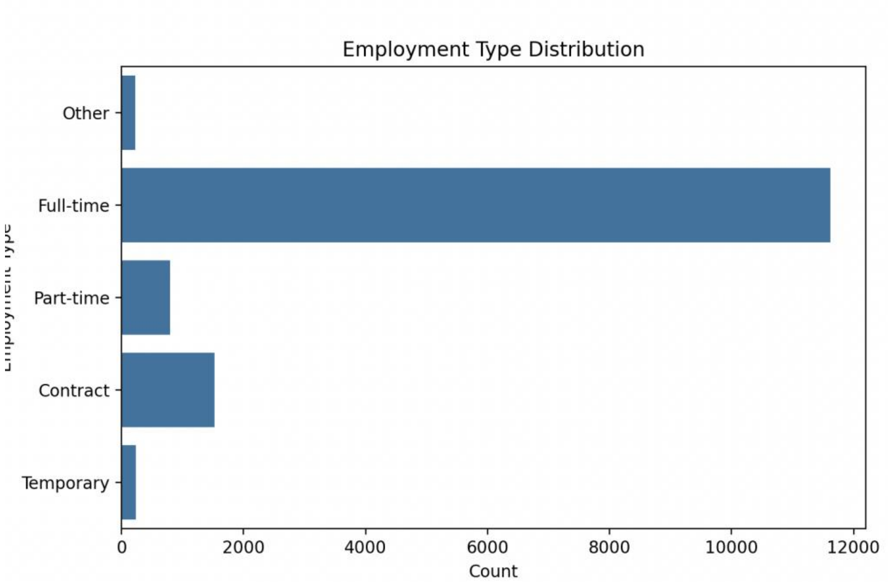
$$Ratio = \frac{State\ \&\ City \mid Fraudulent\ =\ 1}{State\ \&\ City \mid Fraudulent\ =\ 1}$$

Only ratio values greater than or equal to one are plotted below.

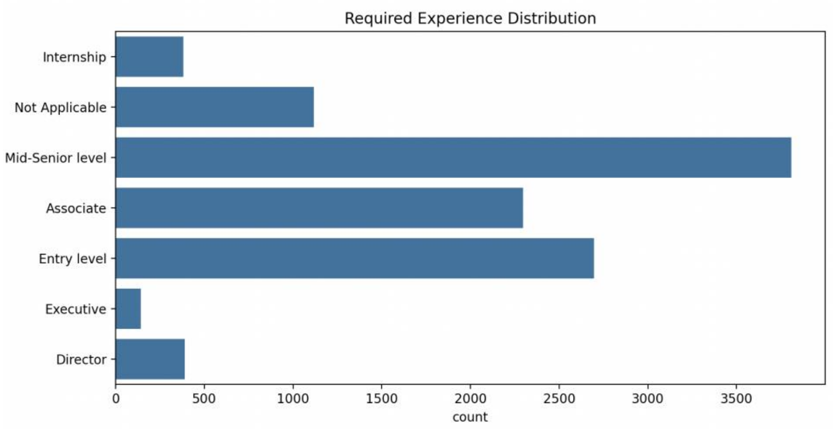


Bakersfield in California has a fake to real job ratio of 15:1 and Dallas, Texas has a ratio of 12:1. Any job postings from these locations will certainly have a high chance of being fraudulent. Other text-based variables are explored further to visualize the presence of any important relationships.

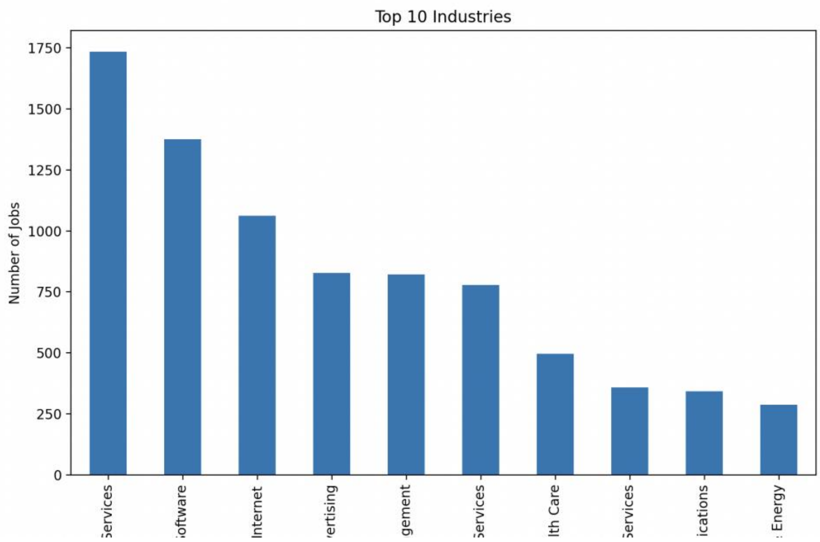
- ➢ Employment Type



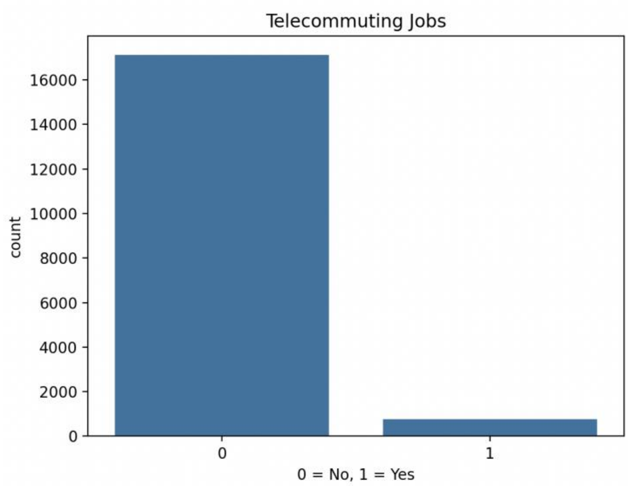
➢ Experience Analysis



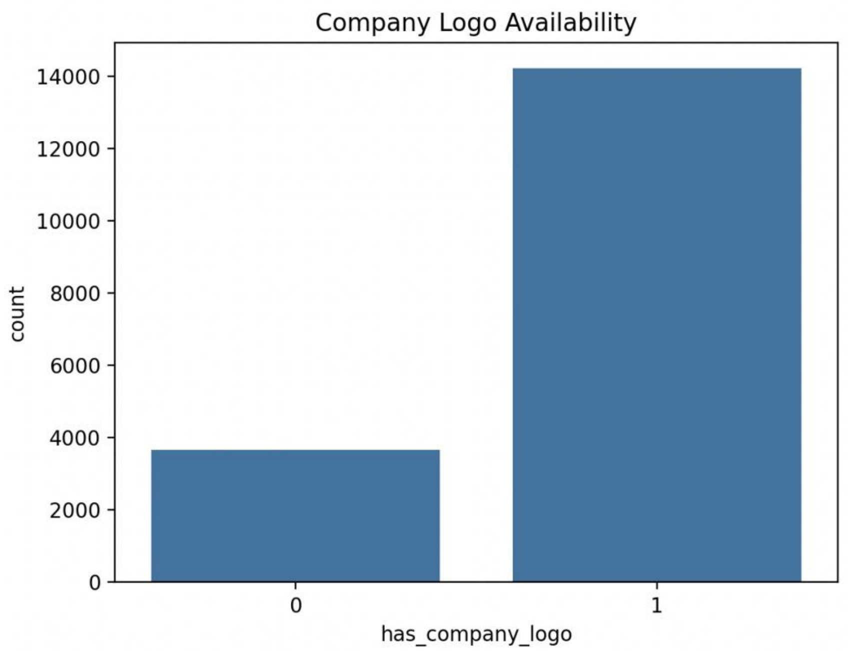
➢ Top Industries



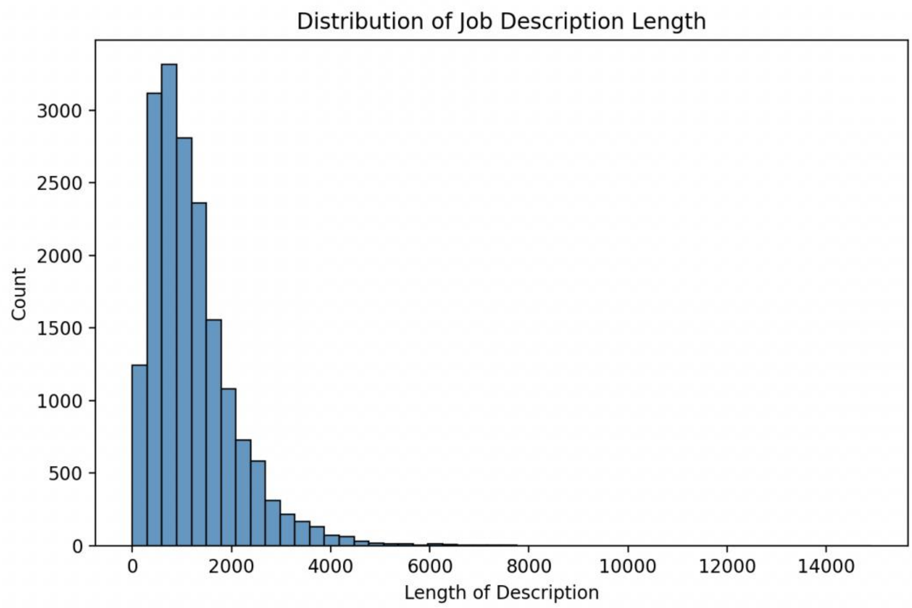
➢ Telecommuting Jobs



- Company Logo Analysis

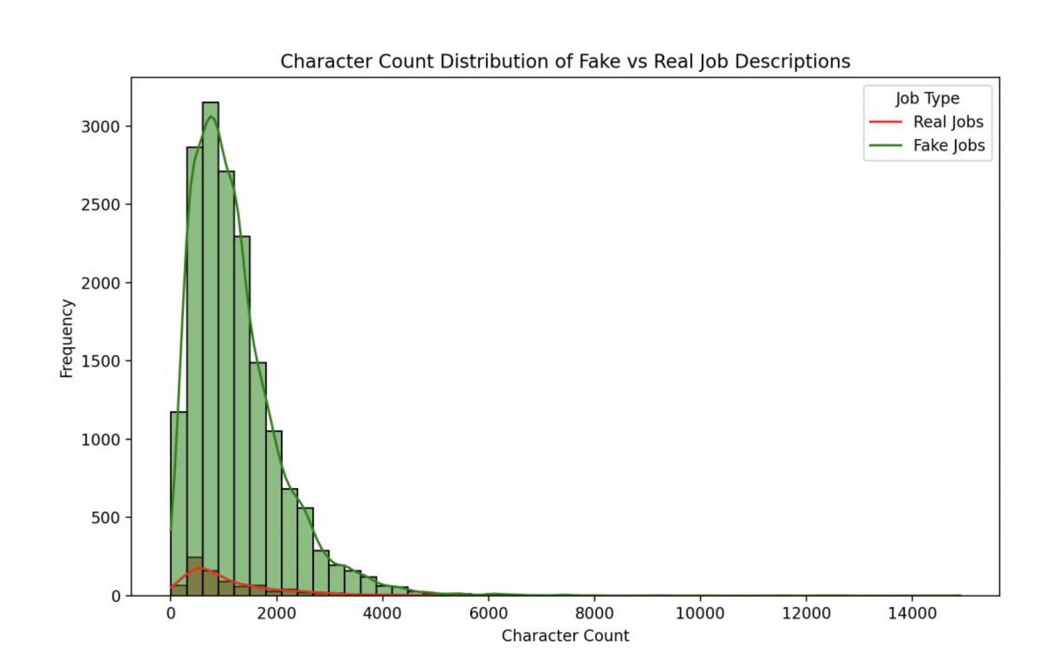


- Job Description Length



The graphs above show that most fraudulent jobs belong to the full-time category and usually for entry-level positions requiring a bachelor's degree or high school education.

To further extend the analysis on text related fields, the text-based categories are combined into one field called text. The fields that are combined are - title, location, company_profile, description, requirements, benefits, required_experience, required_education, industry and function. A histogram describing a character count is explored to visualize the difference between real and fake jobs. What can be seen is that even though the character count is fairly similar for both real and fake jobs, real jobs have a higher frequency.



CHAPTER 4
ALGORITHMS TECHNIQUES AND DATA PREPROCESSING

4.1 Algorithms Techniques

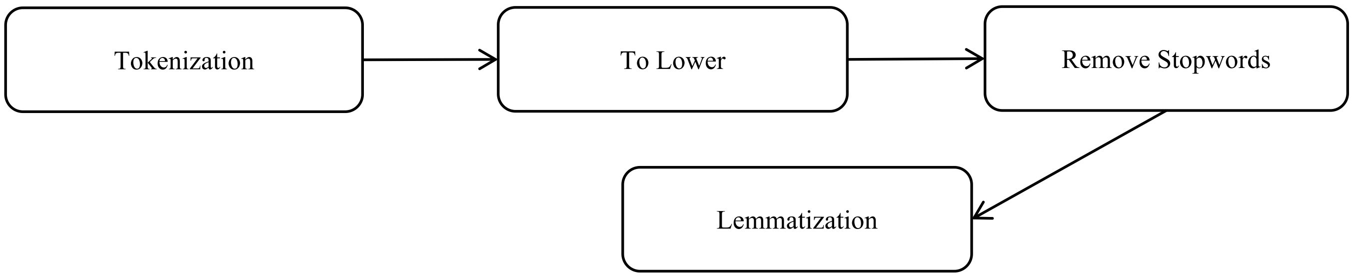
Based on the initial analysis, it is evident that both text and numeric data is to be used for final modeling. Before data modeling a final dataset is determined. This project will use a dataset with these features for the final analysis:

1. telecommuting
2. fraudulent
3. ratio: fake to real job ratio based on location
4. text: combination of title, location, company_profile, description, requirements, benefits, required_experience, required_education, industry and function
5. character_count: Count of words in the textual data Word count histogram Further pre-processing is required before textual data is used for any data modeling.

Further pre-processing is required before textual data is used for any data modeling.

The algorithms and techniques used in project are:
(1). Natural Language Processing
(2). Naive Bayes Algorithms
(3). TF-IDF
(4). Logistics Regression
(5). Train Test Split Techniques
(6). Model Evaluation Techniques
(7). Model Solving Techniques
(8). Web Application Techniques using Streamlit

4.2 Data Preprocessing

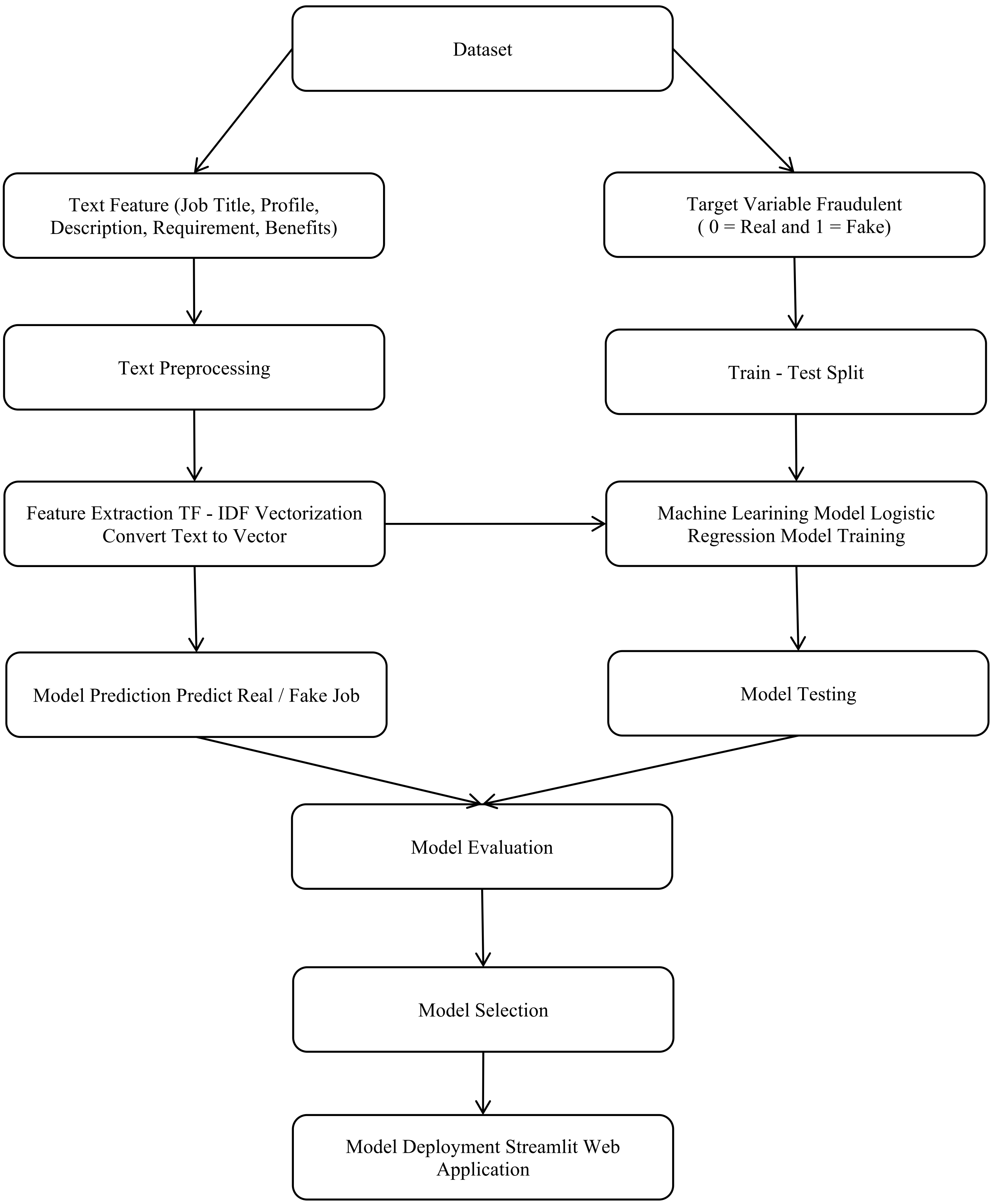


- **Tokenization:** The textual data is split into smaller units. In this case the data is split into words.
- **To Lower:** The split words are converted to lowercase
- **Stopword removal:** Stopwords are words that do not add much meaning to sentences. For example: the, a, an, he, have etc. These words are removed.
- **Lemmatization:** The process of lemmatization groups in which inflected forms of words are used together

CHAPTER 5
MODEL DEPLOYMENT AND IMPLEMENTATION

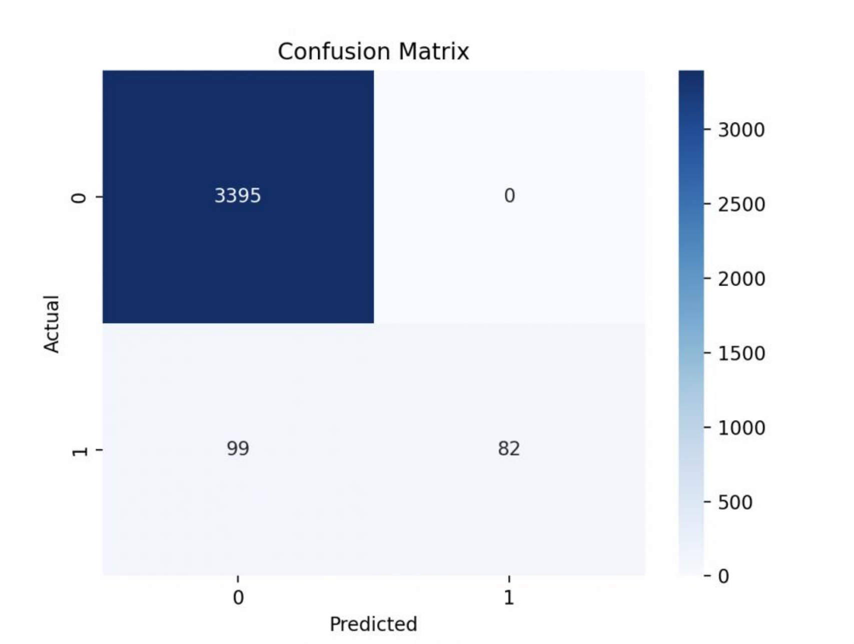
A diagrammatic representation of the implementation of this project is given below. The dataset is split into text, numeric and y-variable. The text dataset is converted into a term-frequency matrix for further analysis. Then using sci-kit learn, the datasets are split into test and train datasets. The baseline model Naive bayes and another model SGD is trained on the using the train set which is 70% of the dataset. The final outcome of the models based on two test sets – numeric and text are combined such that if both models say that a particular data point is not fraudulent only then a job posting is fraudulent. This is done to reduce the bias of Machine Learning algorithms towards majority classes. The trained model is used on the test set to evaluate model performance. The Accuracy and F1-score of the two models – Naive bayes and SGD are compared and the final model for our analysis is selected.

IMPLEMENTATION AND METHODOLOGY DIAGRAM

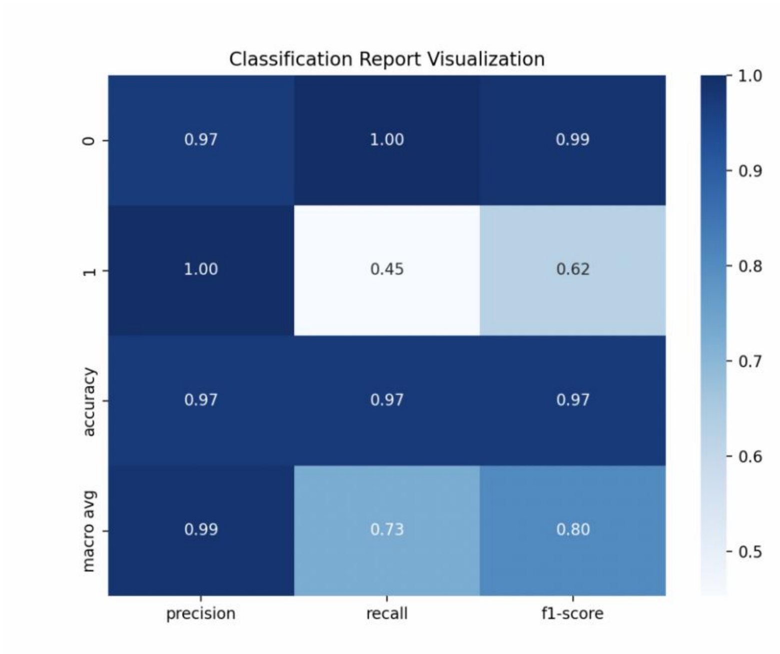


CHAPTER 6
MODEL EVALUATION AND VALIDATION

➢ Confusion matrix



➢ Classification of Report Visualization



CHAPTER 7
CONCLUSION

Fake Job Posting Detection Using Machine Learning

This project focuses on developing a machine learning system to detect fraudulent job postings on online recruitment platforms. With the rapid growth of online job portals, many scammers post fake job advertisements to deceive job seekers by offering unrealistic salaries or requesting personal information and fees. Therefore, detecting fake job postings has become an important task to protect job seekers from online fraud.

In this project, a dataset containing job posting information such as job title, company profile, job description, requirements, and benefits was used. Data preprocessing techniques were applied to clean and prepare the dataset. Natural Language Processing (NLP) methods were used to process textual job descriptions by converting them into a structured format.

The textual data was converted into numerical features using the TF-IDF vectorization technique, which helps identify the importance of words in job descriptions. A Logistic Regression machine learning algorithm was used to train the model to classify job postings as real or fake.

The dataset was divided into training and testing sets to evaluate the performance of the model. The trained model was evaluated using metrics such as accuracy, confusion matrix, and classification report, which helped measure the effectiveness of the model in detecting fraudulent job postings.

Finally, the trained model was saved and integrated into a Streamlit web application, allowing users to input job descriptions and receive predictions about whether the job posting is real or fake.

Overall, the project demonstrates that machine learning and natural language processing techniques can effectively detect fake job postings and help prevent job-related fraud. The system can assist job seekers and recruitment platforms in identifying suspicious job advertisements and improving the reliability of online job portals.